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Complementary Field Effect Transistor (CFET) for the 2-nm Technology Node: A Review From Device to Circuit Perspectives

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ABSTRACT The relentless advancement of the semiconductor industry continues to be fueled by the pursuit of enhanced transistor performance. As CMOS technology undergoes aggressive scaling to enable higher integration densities in modern integrated circuits (ICs), there arises a pressing demand for novel device architectures that can minimize cell area without compromising functionality. The complementary-field-effect-transistor (CFET) is emerging as a leading contender to succeed gate-all-around (GAA) FETs beyond the 2-nm technology node, offering a compelling pathway to sustain Moore's Law into the next era of scaling. Unlike forksheet FET (FSFET) architectures, where nFET and pFET devices are laterally separated, CFETs adopt a vertically stacked configuration. This vertical integration minimizes the required device footprint, constrained primarily by cell height rather than horizontal spacing. As a result, CFETs not only enhance the effective channel width and drive current but also offer the potential for up to a twofold increase in integration density compared to conventional CMOS technologies. This review explores recent advances in CFET architectures, fabrication progress, and associated challenges, exciting prospects in IC design enabled by CFET's compact structure and vertical integration. Finally, the article highlights critical reliability concerns, emphasizing the need for continued innovation to ensure robust device performance. CFET stands at the forefront of next-generation transistor technologies.

INDEX TERMS CMOS, Forksheet, Gate all around (GAA), self heating, SRAM, reliability.

I. INTRODUCTION

For over six decades, the unwavering drive toward 'More Moore'—the continual scaling of transistor dimensions—has been propelled by groundbreaking innovations in device architecture. Pivotal advancements such as fin-field-effect transistor (FinFET) technology, the advent of extreme ultraviolet (EUV) lithography, enhanced interconnect solutions and the integration of high-κ dielectrics have combinedly enabled this remarkable trajectory of miniaturization and performance enhancement [1], [2]. These rapid technological advancements have been vital in enabling the progressive enhancement of integrated circuit (IC) component density, sustaining the momentum of Moore's Law and pushing the boundaries of semiconductor integration. Amidst the mounting difficulties in tackling the short channel effects (SCEs) and the fabricating complexity of FinFETs, GAA field-effect transistor (FET) technology has emerged as the most promising prospect for facilitating transistor scaling at the 2 nm technology node. Among the different setups of the GAAFET architecture, nanosheet-based FET (NSFET) stands out by exhibiting outstanding gate control characteristic, improved tunability and electrical performance. With technology moving beyond the 3 nm node, Complementary FETs (CFETs) have become a very promising architecture,

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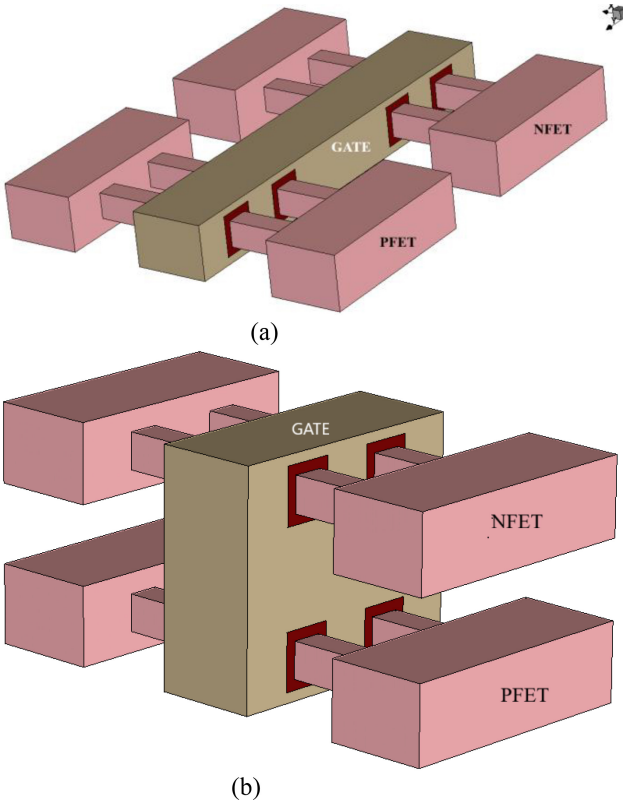


FIGURE 1. **(a) CMOS architecture & (b) CFET architecture.**

which can deliver excellent electrical performance, impressive integration density and reduced parasitic effects [3], [4].

CFETs are exhibiting tremendous attention for their capacity to save active area in the vertical stacking of pFET and nFET devices (Fig. 1). At the same time, they can achieve excellent electrostatic integrity using the GAA or Fin architecture for the channel. Moreover, the vertical stacking of pFET and nFET enables simplified terminal access through internal shorting of the p/n drain contacts, facilitating further reduction in standard cell height and overall chip area [5], [6], [7], [8], [9], [10]. CFET architectures can be broadly categorized as three distinctive architectures: fin-on-rectangular-fin (FORF), fin-on-trapezoidal-fin (FOTF), and fin-on-GAA-fin (FOGF), which have unique structural and performance advantages in the integration of advanced devices [11], [12], [13]. The electrical characteristics of the CFETs are dependent on some key physical parameters including the competencies of the dielectric/semiconductor materials, the p-/n-FET separation thickness (T_{sep}), S/D-extension - the underlap length (L_{un}), Fin height (H_{Fin}) and width of the Fin (W_{Fin}) [14], [15], [16], [17], [18].

CFET technologies can be broadly divided into sequential-CFET and monolithic-CFET architectures relying on the fabrication sequence of the vertically stacked dual layer devices and each has different advantages in integration and process complexity [19], [20], [21], [22], [23], [24], [25]. Monolithic-CFET is a very attractive option, featuring tighter N/P gate integration with no thermal budget-related needs, combined with improved cost-efficiency/wafer, and is a very interesting choice for scaling up of advanced devices [26], [27], [28], [29], [30], [31]. This review emphasizes the recent advances in CFET architectures, process developments & associated challenges, exciting opportunities in circuit design enabled by vertical integration & compact structure of CFET and critical reliability issues and emphasizes the need for further innovation to assure good CFET performance.

II. ADVANCES IN CFET ARCHITECTURES AND PERFORMANCE

The Fin height (H_{Fin}) and Fin-width (W_{Fin}) have big impact on key performance parameters of CFETs for V_{TH} (threshold voltage), DIBL, I_{OFF} (OFF current), I_{ON} (ON current) and SS (swing slope), I_{ON}/I_{OFF}, gate capacitance (C_{GG}) and intrinsic delay (τ_{delay}) [32], [33], [34], [35], [36]. The decreased W_{Fin} results in the increase in V_{TH} and decreased SS & DIBL for CFETs. It also minimizes the I_{OFF}, and hence gives a great boost to the I_{ON}/I_{Off} ratio. However, this comes at the cost of degraded I_{ON}. As W_{Fin} increases, C_{GG} slightly rises due to the larger gate area, and the τ_{delay} decreases because of the enhanced I_{ON}. Therefore, decreasing W_{Fin} presents a trade-off between improved gate control and reduced drive current. As the H_{Fin} shrinks, the narrow width effect becomes more pronounced, leading to a rise in V_{TH} and enhanced electrostatic control, reflected by reductions in both DIBL & SS. On the AC traits, the total C_{GG} experiences a notable decline due to the diminished gate and S/D sidewall area. While this reduction in C_{GG} favors lower dynamic power consumption, it is partially offset by an increase in τ_{delay}, driven by the degradation in I_{ON} resulting from a reduced effective channel width [11], [37], [38], [39], [40], [41], [42]. To boost the transconductance (g_m) and I_{ON} of CFETs, tree-shaped channel having vertically stacked nanosheet (NS) channels interconnected by SiGe-Fin-inter-bridges (IB) can be adopted (Fig. 2).

The I_{ON} for both N-Tree and P-Tree CFETs enhances with rising H_{IB}, primarily due to the enlarged channel area (Fig. 3 (a)). In contrast, the I_{OFF} remains relatively unchanged. As a result, the I_{ON}/I_{OFF} ratio improves with higher H_{IB} (Fig. 3 (b)). Additionally, as H_{IB} increases, the SS of the N-Tree CFET remains relatively stable, while the SS of the P-Tree CFET shows a decreasing trend (Fig. 3 (c)). Meanwhile, the DIBL for both N-Tree and P-Tree CFETs declines consistently (Fig. 3 (d)). This trend can be explained by the fact that as H_{IB} becomes substantially larger than inter-bridge width (W_{IB}), the potential degradation in gate control at the NS-IB interface is effectively compensated by the improved gate control enabled by the increased H_{IB}. The C_{GG} increases with increasing H_{IB}, and this can be attributed to the increased channel area essentially, which is associated with a higher oxide capacitance (Cox) (Fig. 3(e)). In addition, the fin height increases as a whole with H_{IB}, thus increasing the fin overlapping between the S/D and the gate electrodes, and thereby the parasitic capacitance. As H_{IB}

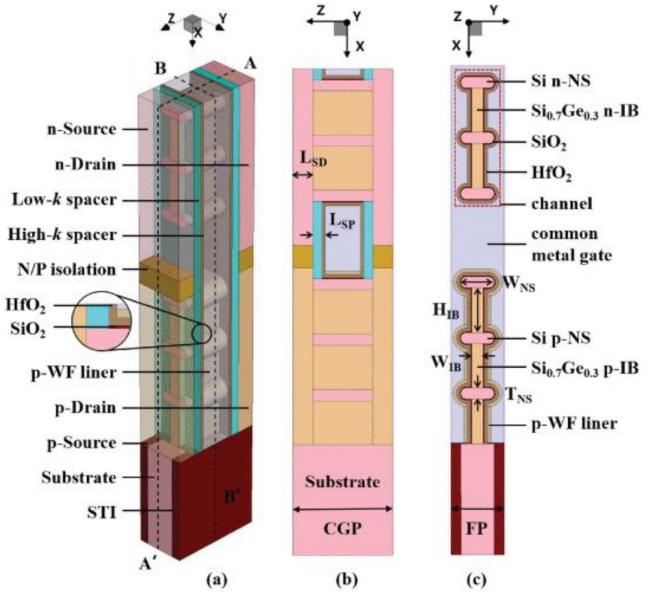


FIGURE 2. CFET with tree-shaped nanosheet channels [43] (Copyright@IEEE).

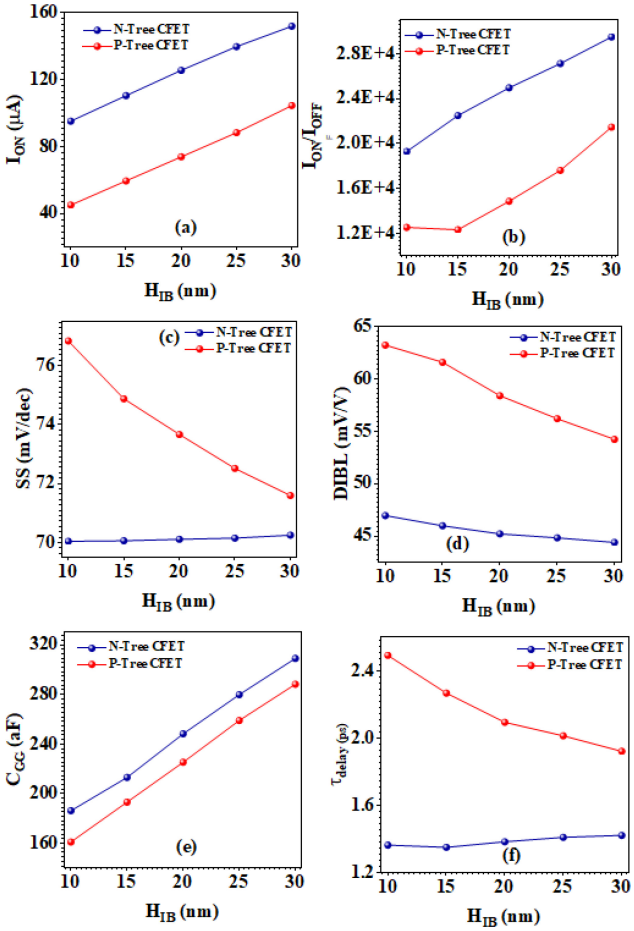


FIGURE 3. Impact of H_{IB} on (a) I_{ON} (b) I_{ON}/I_{OFF} (c) SS (d) DIBL (e) C_{GG} and (f) τ_{delay} of CFETs [43].

increases, the intrinsic time constant (τ_{delay}) of the N-Tree CFET shows a general increasing trend, while that of the P-Tree CFET shows a continuous decrease (Fig. 3(f)). These behaviors are attributed to a combination of the influence of changes in I_{ON} and C_{GG}. As the amount of W_{IB} increases, both the I_{ON} in the N-Tree CFET initially rises and then falls and the I_{ON} in the P-Tree CFET rises continuously but at a reduced rate. These trends are probably limited by the limitations of structural device design. Concurrently, the I_{ON}/I_{OFF} ratio decreases steadily with an increase in W_{IB}, which suggests a decline in switching performance. This is the result of the increase in carrier density and leakage current for the inner branches (IBs) as W_{IB} expands. Moreover, both the SS and DIBL increase with more W_{IB}, which can be explained by the wider interface between NSs and IBs, resulting in reduced gate control. The C_{GG} is reduced monotonously by the reduced overlapping area between the S/D and the gate metal electrode, resulting in reduced parasitic capacitance. As a result of the accumulation of increased I_{ON} and a reduction in C_{GG} the delay is reduced with increasing W_{IB} [43].

3D stacking of transistors is a promising way to continue scaling of devices within a certain fixed footprint. Among the different architectures, the CFET has attracted a lot of attention regarding the possibility of integration density improvement. However, monolithic Integration of the middle dielectric isolation (MDI) process, achieving vertical isolation between the S/D contact metals of the top and bottom devices, and Epitaxial growth of differently doped layers for the top and bottom transistors are some of the major challenges for monolithic integration of CFETs. These problems greatly complicate the process and require a great deal of research and innovative ideas to tackle before the technology can be implemented on a large scale. Alternatively, vertical field-effect transistors (VFETs) have shown strong prospect as the 3D integrated architecture owing to their structural flexibility, high integration density and excellent electrostatic control. Compared to traditional lateral based field effect transistors (FETs), VFETs are characterized by a reduced set of constraints on the gate length and the S/D contact area, which are due to their vertical integration approach. This makes it possible to create more compact layouts as well as better electrostatic performance [44].

Incorporation of a single crystalline ferroelectric (FE) layer with a stable negative capacitance (NC) into the gate stack of CFETs is one of the promising routes to suppress SCEs, decrease SS, reduce effective oxide thickness (EOT), and reduce operating voltage [45], [46], [47], [48]. Complementary Ferroelectric FETs (CFeFETs) have good prospects in next-generation computation-in-memory and non-volatile memory (NVM) applications [49], [50]. Complementary organic field-effect transistors (COFETs) are ideal candidates for becoming vertically integrated on flexible substrates, thanks to their ability to be solution

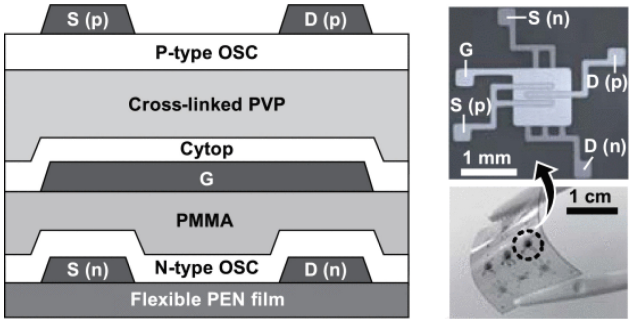


FIGURE 4. Flexible organic CFET [51] (Copyright@IEEE).

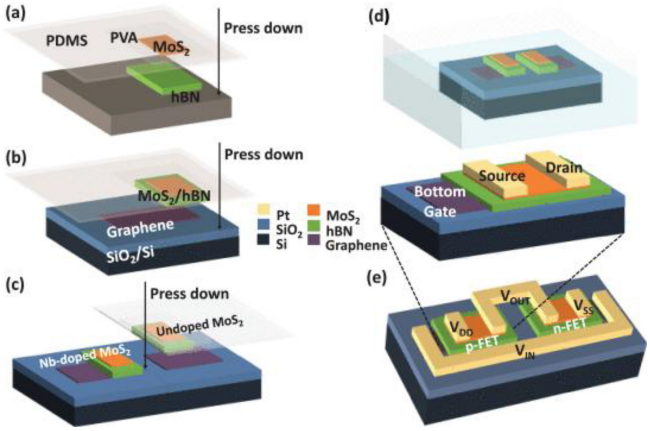


FIGURE 5. Fabrication procedure of 2D-materials based CFET [52] (Copyright@IEEE).

processed layer by layer deposition at low temperature. In addition, 3D stacking is beneficial to mitigate the number of interconnection paths between transistors, paving the way for more complex electronic circuits [51]. Vertical stacking of thin-film transistors (TFTs) is an excellent approach to lower the footprint of a device, hence boosting the density of FETs in complicated flexible electronic applications without lowering the feature size and resolution of the patterning tools. Kyung et al. [51] successfully fabricated a high performance, solution processed organic CFET using polymer semiconductors (P-24-PNDI-TVT (n-type) & P-29-DPP-SVS (p-type)) on PEN substrate (Fig. 4). This 3-D organic flexible CFET inverter demonstrated an excellent noise immunity of up to 60% of $V_{DD}/2$ and a peak static voltage gain of ≈18 V/V. The adoption of solution-processable dielectric materials with high chemical resistance, outstanding dielectric traits, and compatibility with organic semiconductors is crucial in the manufacturing of 3-D stacked organic transistors. These critical factors allow for the use of dielectrics like PMMA for the NFET and Cytop/cross-linked PVP bilayer for the PFET in flexible organic CFETs. Flexible Organic FETs with PVP dielectric-films often showcase a large gate leakage and significant hysteresis. By cross-linking the PVP films with agents like PMF, EAD, HDA, and HMBG, these undesirable traits can be prevented.

CMOS- circuits based on 2D-FETs have opened up new horizons to advance the IC technologies of modern times. Beyond solving the scaling limitations of the conventional FETs, these devices have exceptional promise for enabling the next generation of flexible electronics that will power innovations such as foldable smartphones, wearable electronics and curved display technologies. Among the various 2D materials, MoS_2 has been extensively investigated and has demonstrated outstanding n-channel transistor performance. The manufacturing process flow of 2D-material MoS_2) based CFET is demonstrated in Fig. 5. The performance of 2D-CFETs is often constrained by high S/D contact resistance (R_c), making it essential to reduce the Schottky-barrier-height (SBH) at the 2D-semiconductor/metal interface [53]. Usually, WSe_2 and MoS_2 are adopted as p-type and n-type channel materials, respectively, in 2D-CFETs [54]. In the era of NSFETs, the semiconductor industry has recognized 2D material as a viable replacement for the traditional Si channel.

TABLE 1. Comparison of NSFET, FSFET & CFET inverter designs.

Parameter	NSFET	FSFET	CFET
Electrostatic control	Low	Moderate	High
On Current (I_{ON})	Moderate	High	Moderate
Off Current (I_{OFF})	Low	Low	Moderate
Parasitic capacitance	Moderate	Low	Moderate
Manufacturability	High	Moderate	Low
Area Efficiency	Low	Moderate	High
Circuit Delay	High	Moderate	Low
Thermal Concerns/Self heating effects	Low	Moderate	High
Variability sensitivity	Moderate	Moderate	High
Cell Height Scaling	Low	Moderate	High
Parasitic Coupling	Low	Moderate	High
Routing Complexity	High	Moderate	Low

Among various GAA technologies such as NWFET, NSFET and FSFET, CFET is renowned for its substantial area efficiency. Through vertical stacking of pFET and nFET GAA devices, CFET design minimizes lateral constraints on space, lowering the silicon footprint by approximately 50% compared with standard GAA designs such as the Nanosheet FETs & Forksheet FETs. In addition to this compact architecture, CFET retains outstanding performance as the nFET and pFET devices incorporated in CFETs retain the advantages of GAA nanosheet structures. The performance comparison of NSFET, FSFET & CFET inverter designs is given in Table 1 and the benchmarking of CFET architectures is given in Table 2.

III. CFET FABRICATION TECHNOLOGY & CHALLENGES

Fig. 6 illustrates a monolithic process flow that begins on a bulk silicon wafer. The wafer is then doped with a dopant type opposite to that of the bottom FET to enable the formation of an isolation junction. An epitaxial stack

TABLE 2. Benchmarking of CFET architectures.

Ref	Architecture	L_G (μm)	V_{TN} (V)	V_{TP} (V)	V_{DD} (V)	Gain	I_{ON}/I_{OFF}
[52]	2D-CFET	2	1.5	-0.5	2	14	10^7
[51]	Organic-CFET	35	7.4	-5.6	30	18	4.4×10^7
[43]	Tree-CFET	0.016	-	-	0.7	-	$2.49X10^4$
[44]	Vertical-CFET	0.065	0.56	-0.42	0.8	13	5.4×10^6
[47]	NC-CFET	0.15	0.85	-0.4	1.2	30	$3.5x10^5$
[62]	Ge-NW-CFET	0.1	0.88	-0.14	1	70.5	-
[60]	Hetero. CFET	0.06	0.26	-0.61	1.25	21	-
[14]	CFET	0.080	-	-	0.7	10.4	10^6

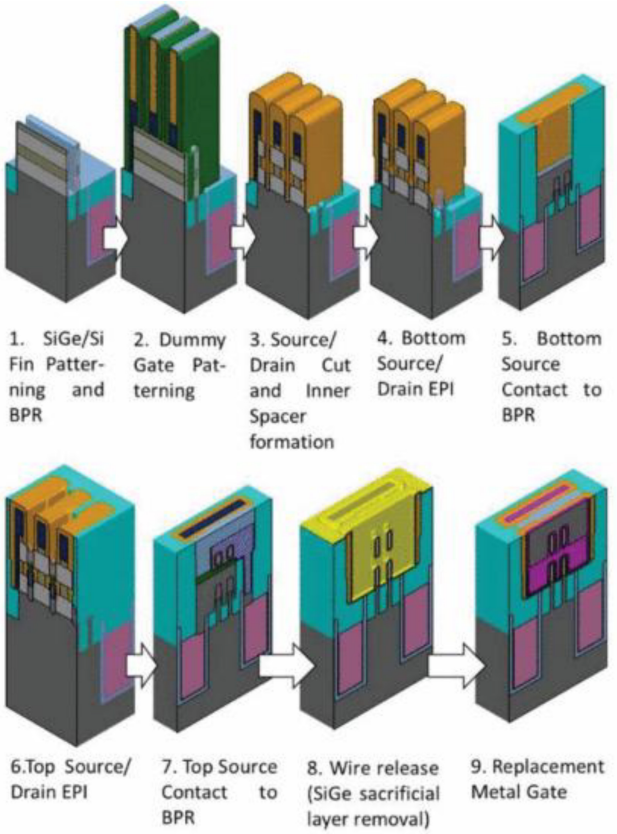


FIGURE 6. Monolithic CFET process flow [56] (Copyright@2020IEEE).

consisting of a SiGe sacrificial interlayer, Si-top channel, SiGe-cap, and Si bottom channel is grown on the silicon wafer. Fins are then patterned down to the wafer and buried power rails (BPR) are subsequently formed. Subsequently, dummy gates can be patterned, and the Si/SiGe stack in the prospective S/D regions can be selectively etched. An inner spacer should be introduced between the two channel layers to ensure proper isolation. To enable S/D epi-growth on the bottom Si-channel while preserving the integrity of the top FET, a Silicon-On-Carbide shielding layer can be deposited, followed by spacer formation along the sidewalls of the top device. The bottom S/D connection to the BPR lines can be achieved by first sequential patterning contact trenches and vias, and then metallized with cobalt (Co). Then, selective isolation layer can be developed on top of the bottom metal, for the epi-growth of the S/D of the top-FET. Similar to the bottom level, the top S/D regions are connected to the BPR through patterned contact trenches and vias, which are subsequently filled to complete the electrical pathway. The bottom S/D is connected to the BPR lines through sequential patterning of contact trenches and vias, followed by cobalt (Co)-metallization. A selective layer of isolation is then grown over the bottom-metal to allow for S/D epitaxial growth on the top FET. Subsequently, the top S/D regions are also connected to the BPR using a similar process of contact trench and via patterning, followed by metallization. Finally, the gate metal replacement module is completed by simultaneously removing the sacrificial-SiGe and implementing GAA-deposition for the top-FET, while a top-cladding-gate is formed for the bottom-FET. This process flow consists of a high level of fabrication complexity including advanced process steps such as selective dielectric deposition onto metal surfaces, precision recessing of material, and filling of metals into buried-structures [57].

Amorphous-Si (α-Si) can be deposited on SiO_2 using LPCVD technique. Solid-phase crystallization (SPC) method can be adopted for recrystalizing α-Si for 24 hours at 600°C. Then, the Si-bottom-layer will be implanted with BF_2-ions for JL-P-NSFET and ^{31}P-ions for JL N-NSFET. Following furnace-annealing at 600 °C for 4 hours to activate the doping ions, the Silicon was subsequently thinned using dry-etching. An interlayer of SiO_2 can then be deposited using PECVD for isolation. The top α-Si layer can then be fabricated in the same way as the bottom α-Si layer (Fig. 7). The S/D regions of the top-NSFET need to be partially etched to create a window for S/D-contact regions of the bottom NSFET. After this, gate patterning, dielectric passivation and contact metallization will be done for CFET fabrication. In 2020, Subramanian et al. [7] reported the first-ever monolithic integration (MI) of 3D-CFET on 300 mm substrates. The monolithic CFET process offers a more cost-effective solution than the sequential CFET process. Unlike the sequential 3D process, the monolithic 3D approach does not require layer-transfer or wafer bonding. Additionally, monolithic 3D integration via epitaxy can lower both FEOL costs and parasitic RC compared to the sequential 3D method [58], [59], [60]. In 2021, Hong et al. [8] reported the fabrication of heterogeneous CFET with Si and Ge channels using layer-transfer-process. The 3D-nanosheet-channel stacking-integration specifically utilizes a low-temperature (below 200 °C) hetero-layer-bonding-technique (LT-HBT), achieved through surface-activating chemical-treatment at room-temperature, allowing Ge-channels to be bonded onto Si wafers.

In 2024, Hsieh et al. [60] demonstrated monolithic 3-D stacked GeSi and GeSn self-aligned CFETs with more P/N-junction-isolation by adopting in-situ doped CVD-epitaxy which can effectively minimize the complexity of monolithic CFET fabrication. Multiple P/N-junctions are employed for isolation, replacing the dielectric isolation between the bottom and top NSFETs. Sacrificial layers (SLs) with heavy doping help reduce S/D resistance through thermal annealing during CFET manufacturing, eliminating the requirement

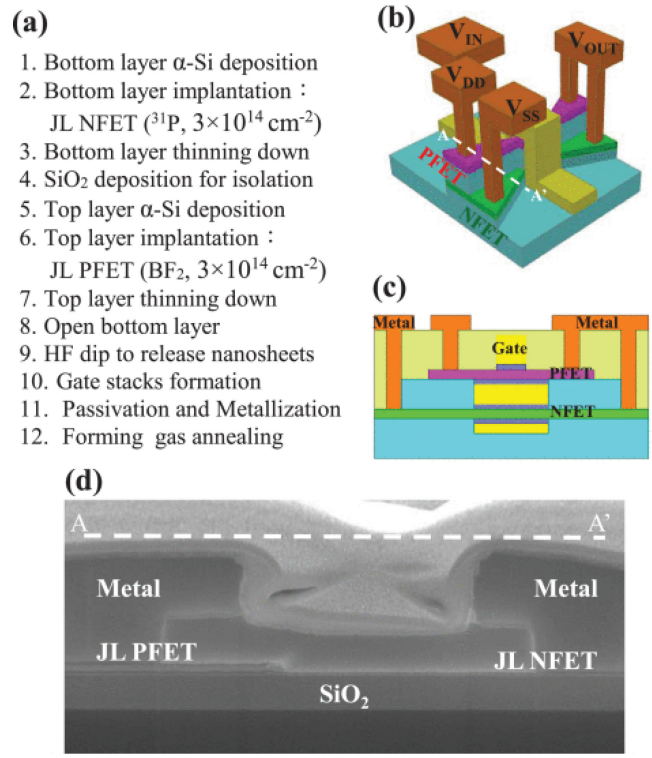


FIGURE 7. Process flow for a vertically-stacked CFET inverter [57] (Copyright@2020IEEE).

for S/D selective epitaxial regrowth in this architecture. Additionally, the heterogeneous dual-channel design enables V_{TH} matching by aligning the energy bands of the individual channels, using a single work function metal (WFM) for the common gate architecture. At the initial phase of the epitaxy process flow, the top silicon-layer of the SOI wafer is thinned via thermal oxidation in a vertical furnace, followed by buffered oxide etchant (BOE) dipping. After wafer preparation, epitaxial layers are grown using rapid thermal (RT) CVD, preceded by an HF dip to remove the native oxide. Following a 1100 °C H_2-bake to further eliminate the native oxide, a thick undoped Ge-buffer layer can be deposited, accompanied by in-situ annealing at 800 °C in an H_2-ambient for 10 minutes. This annealing step confines misfit dislocations at the Si/Ge interface, enabling subsequent good-quality epi-growth. For the bottom n-NSFET, the Ge:P SL and a single GeSi-channel layer can be grown. Electrical-isolation between the bottom and top FETs can be achieved by growing the Ge:B and Ge:P-layers in three repeated cycles. Following the epitaxial growth of Ge:B/Ge:P-isolation-layers, stacked B-doped GeSn channels, separated by Ge-SLs, can be grown for the top p-NSFETs. The B-doping in the GeSn-channels improves the etching selectivity of Ge-SLs over the GeSn-channels, facilitating channel release via wet etching.

Following CVD epitaxy, a SiO_2 hard mask is deposited via PECVD to prevent direct exposure of the epitaxial layers to organic photoresist. The channels and S/D regions are then patterned using e-beam lithography, and fin structures are formed through Cl_2-based reactive ion etching (RIE). Subsequently, a field oxide is deposited by PECVD to shield the channels from oxidation and shield the S/D regions during the channel release process. Channel release is achieved through wet etching using an NH_4OH and H_2O_2 mixture, simultaneously forming the GeSn and GeSi nanosheets. After release, $HfZrO_2$ gate dielectrics are conformally-deposited around the nanosheets via PEALD, followed by in-situ TiN deposition and 400 °C forming gas annealing. A thick TiN layer is then deposited via PVD to serve as the gate metal pad. Following gate-stack formation and gate metal pad patterning, the epilayers of the top p-NSFET and the P/N junction isolation in the S/D regions are partially etched through a deep step formation process, providing access to the bottom n-NSFET's S/D regions. Finally, S/D contacts can be deposited using PVD Pt, followed by a 400 °C PMA to form PtGe, thereby reducing S/D resistance. Typically, CFETs involve one p-channel, and one n-channel, which makes the process simple & straightforward. In contrast, multi p-SiGe and n-Si channels in the fabrication of CFETs are more difficult because these structures need two opposing selective etching processes, namely removal of Si for the p-NSFET and retention of SiGe and removal of SiGe for the n-NSFET and retention of Si [60], [61], [62], [63], [64]. Typically, CFETs incorporate a single p-channel and a single n-channel, making the process relatively straightforward. In contrast, fabricating CFETs with multiple p-SiGe and n-Si channels is more challenging, as it requires two opposing selective etching steps—removing the Si layer while retaining SiGe for the p-NSFET, and removing the SiGe layer while retaining Si for the n-NSFET [60], [61], [62], [63], [64].

IV. INTEGRATING CFETS IN ADVANCED CIRCUIT DESIGN

In each generation of advancing technology, the integration of on-die static-random-access-memory (SRAM) (Fig. 8) is progressively scaled up, which is particularly the driver in boosting the performance of system-on-chips (SoCs) [65], [66], [67], [68], [69], [70]. This is facilitated by the shrinking of SRAM-bitcell-area with technology. Data centric applications with high energy efficiency requirements impose stringent requirements on large capacity embedded SRAM arrays which are able to operate with low voltages (V_{DD}) [70], [71], [72], [73], [74]. Achieving a large memory capacity with a minimum of cell area is crucial to increase computing capabilities. CFET technology exhibits extraordinary scalability, and has been proven to continue the SRAM scaling pace beyond the trajectory of NSFETs. Nevertheless, 6T CFET SRAM still suffers from read/write conflict and requires the use of assist circuits. In 2022, Lee et al. [68] reported a 8T-CFET$_{BEOL}$ SRAM for the first time which achieved better performance results in terms of energy efficiency, stability and high speed than 6T-CFET SRAMs. Monolithic 3D stacking of CFET SRAM arrays allows for multifold higher integration density as well while

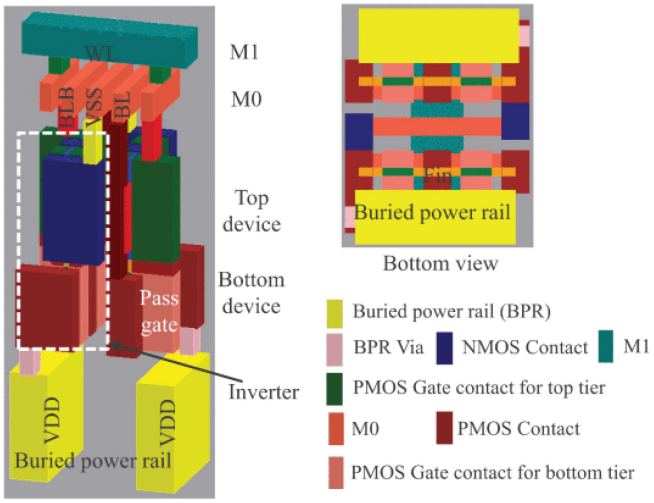


FIGURE 8. CFET-based SRAM: 3D-architecture [66] (Copyright@2021IEEE).

retaining all the innate advantages of SRAM such as ultra-low power for writing and greater endurance. By integrating CFET and BPR technology, leveraging vertical contact and routing resources can result in more than a 40% reduction of cell area. In 2023, Ding et al. [71] proposed an innovative design of the zigzag-cell design for the 6T-SRAM bitcell in the CFET framework. This zigzag cell design retains the aggressive area reduction while allowing the bitlines to be shorter by taking advantage of the circuit design for CFET architecture.

In 2023, You et al. [72] presented a single-ended write-enhanced 11T CFET SRAM, thus realizing the first demonstration of reconfigurable compute-in-memory (CIM) function technology in a single bit cell. By using the dummy p-FETs in the conventional 6T (or 8T) CFET SRAM layout, this innovative 11T SRAM realizes more than 2.5 times higher write ability than conventional 6T (or 8T) SRAM, without sacrificing write half-select disturb immunity. Furthermore, the dummy p-FETs, functioning as supplementary write transistors, enable the realization of Boolean CIM with three distinct reconfigurable schemes. In 2024, Cheng et al. [74] reported an innovative design of high-density, ultra-scaled 1-nm node lateral CFET-based 6T-SRAM which can substantially reduce the layout area. In advanced SoCs, multi-port embedded memories have become one of the important enablers of higher performance, and enable simultaneous write and read operations to greatly increase the overall system efficiency. Abdi et al. [75] demonstrated a two-port CFET based 8T-SRAM bit cell which achieved 53% reduction in the area footprint when compared to the traditional NSFET-based 8T-SRAM bit cell. Digital compute-in-memory (DCIM) has recently burst to the scene as an attractive alternative to overcome the scalability and accuracy limitations posed by analogue CIM (ACIM), a future accelerator of AI hardware. Shon et al. [76] demonstrated a 3D-DCIM accelerator based on CFET technology which achieves 70% better FOM than conventional 1-nm DCIM design.

While CFET SRAM has significant advantages in terms of cell area reduction and higher overall density, it presents significant challenges in the routing of signal and power lines in the ever-shrinking area. As the cell size keeps shrinking, the available routing space particularly in the back-end-of-line (BEOL) layers above the SRAM cells is increasingly constrained [75], [76], [77], [78], [79]. Traditionally, the routes of the wires are taken in the BEOL so that they lie on top of the devices. However, with continued device density in the front-end-of-line (FEOL), the routing space in the BEOL continues to dwindle. This limitation requires very smaller wire widths which results in increased parasitic resistance and thus degraded signal integrity and power delivery. Moreover, the lower spacing increases the parasitic capacitance situation among the wires as their higher proximity increases. The added parasitic capacitance and resistance degrade the signal further which may, in turn, slow down both the read and write operations of the SRAM. In SRAMs, a double-sided signal/power routing architecture proves significantly more effective than single-sided signal routing with BS-PDN, particularly for energy-efficient, data-centric AI edge applications [78].

V. PROCESS VARIATION EFFECTS & RELIABILITY ISSUES

Process fluctuations pose a significant challenge to the reliable design of integrated-circuits (ICs) at advanced technology nodes. Much like FinFETs and GAA-FETs, CFETs operating at the 3-nm node and beyond are inherently susceptible to both intrinsic device-level random-fluctuations and systematic process variations [80], [81], [82], [83], [84]. The important sources of random variability include oxygen (O_2) vacancies in gate-dielectric, random telegraphic noise (RTN), line edge roughness (LER), random dopant fluctuation (RDF), oxide thickness variation (OTV), gate edge roughness (GER), intrinsic metal gate granularity (MGG)-induced work-function variability (WFV), and so on (Fig. 9). In 2022, Yang et al. [81] investigated the process fluctuation (WFV, LER, GER & RDF) effects on DC characteristics of CFETs for the first time. In comparison to traditional GAA-NSFETs, the primary distinction in CFETs lies in GER. The variation in electrostatic potential caused by GER in CFETs is influenced not only by the shared metal gate but also by the extra p-type work-function (p-WF) liner used for p-FETs. Consequently, GER exerts a much stronger impact on p-FETs than on n-FETs or GAA-NSFETs. To mitigate this drawback, increasing the thickness of the p-WF liner has been proposed. V_{TH} and I_{OFF} of CFETs are observed to be more sensitive to WFV than I_{ON} since the gate metal WF directly affects the WF difference between the semiconductor and metal. LER manifests along both the nanosheet width and height directions, referred to as line-width roughness (LWR) and line-height roughness (LHR), respectively. LHR primarily arises from variations during Si/SiGe layer deposition and SiGe release etching, whereas

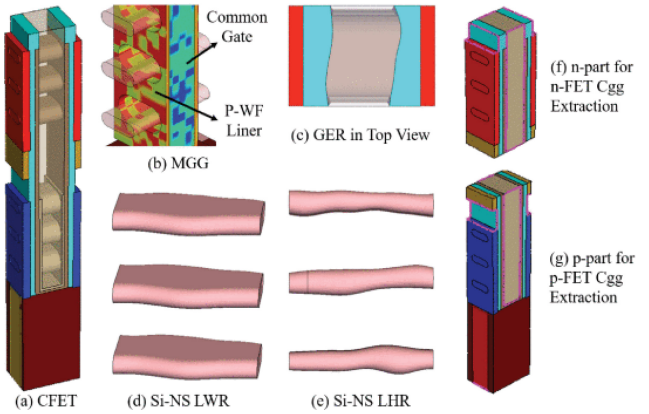


FIGURE 9. (a) 3D-view of CFET (b) MGG (c) GER (d) LWR (e) LHR (f) n-FET C_{gg} (g) p-FET C_{gg} [82] (Copyright@2023IEEE).

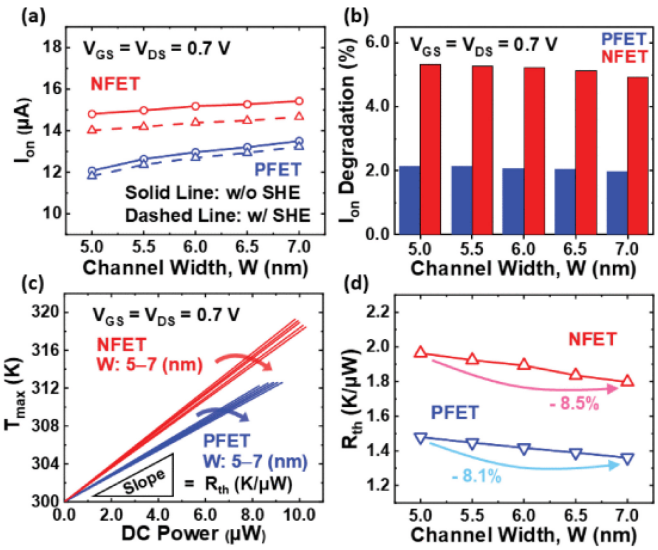


FIGURE 10. (a) I_{on} vs W with and without SHE (b) I_{on} degradation rate vs W (c) T_{max} vs DC power and (d) R_{th} vs W [86].

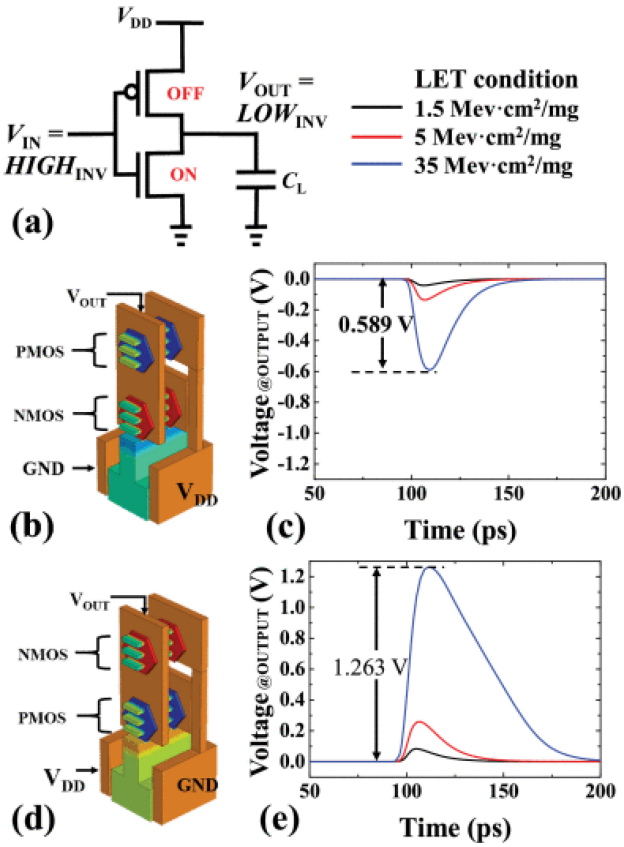


FIGURE 11. Heavy ion strike effects in CFET based inverter [96] (Copyright@2025IEEE).

LWR is predominantly induced by lithographic limitations. LER and GER occur along different orientations: LER extends along the channel width, whereas GER propagates along the channel length. Compared to WFV and LER, the variation induced by GER is relatively minor, as the spatial extent of GER-induced fluctuations along the channel is limited. RDF is influenced by the doping concentration in the channel. RDF is negligible in lightly doped channels. In lightly doped CFETs, RDF in the S/D regions plays a dominant role in performance variability. When RDF is present in both the channel and S/D regions, a reduction in channel depth (DCH) diminishes the channel's contribution, thereby significantly lowering the overall variability. Furthermore, decreasing DCH slightly increases the doping concentration in the S/D regions due to impurity compensation effect.

CFETs integrated with BPR are emerging as a promising solution for sub-3 nm technology nodes. However, in such 3D-stacked architectures, the compact topology exacerbates the self-heating effect (SHE) by limiting phonon scattering. This intensified SHE is a critical reliability issue which enhances degradation mechanisms such as the bias temperature instability (BTI) and hot carrier injection (HCI) leading to a reduced device lifetime [85], [86], [87], [88], [89], [90], [91]. In aggressively scaled devices, increasing short-channel leakage currents tend to aggravate the damage to devices, in the form of heat, and thereby limit their applications to both dense integration and high-performance applications. The channel width (W) is a crucial geometrical parameter for optimizing the DC performance, which has great effects on the DC performance and the SHE of multi-gate transistors.

Increasing the W causes an increase in the on-current and reduces RC delay & I_{ON}/I_{OFF} of both NFET and PFET, as a result of the bigger effective channel area (Fig. 10(a)). However, Fig. 10(b) shows that the SHE produces charge scattering and mobility-degradation resulting in a decrease in on-current. Notably, the on-current degradation in NFET is more pronounced than that in PFET, which is attributed to the higher lattice temperatures. In addition, Fig. 10(c) shows that the higher the W, the higher the maximum lattice temperature Tmax caused by the higher DC power dissipation and thus the stronger Joule heating. The slope of Fig. 10(c) gives the thermal resistance (Rth) characterizing the correlation between power consumption and temperature

rise; a low value of Rth represents a good heat dispersion and improved SHE management. As shown in Fig. 10(d) the top NFET is more vulnerable to SHE than the bottom PFET. This increased susceptibility is the result of the reduction of thermal conductivity of the NFET and because the channel is completely encapsulated with dielectric material, limiting heat flow. Importantly, it was seen that increasing W significantly reduces Rth in both the FETs because that increase the transfer of heat through the S/D regions and the underlying silicon bulk in between by increasing the width of the active region. Consequently, the lowered Rth is responsible for the decrease of on-current degradation rate for increasing W. These results indicate that it is important to carefully optimize W to improve the heat dissipation and reduce the SHE of CFETs.

Electromigration (EM) in metal interconnects is a constant reliability problem especially for above the 3 nm node, of particular concern in the case of BPRs as stacked device architectures [91]. The aggressive scaling of interconnect dimensions dramatically increases the current density, creating a higher stress in metal lines. At the same time, the decreasing metal pitch causes smaller grain sizes (GSs) to appear so many grain boundaries are created from the bottom and sidewalls from the advanced nodes. These grain boundaries can provide additional diffusion routes for metal ions and so increase EM failures in interconnects. At present, ruthenium (Ru) and cobalt (Co) are the most popular candidates to replace copper (Cu) in the production of advanced metal interconnects. With continued operation of device scaling, the trap density in the gate dielectric is increased, increasing the reliability issues and magnifying the effect of trap-induced noise [92].

The booming industry of space requires highly robust electronics in order to ensure the safety as well as efficiency of the space missions. One of the most important challenges comes from the space environment which is harsh and includes radiation from heavy ions, protons, electrons and various high-energy electromagnetic waves like X-rays and g-rays, leading to serious deterioration of device performances. In particular, the cumulative damage over time is significant with total ionizing dose (TID) irradiation. Hence, the understanding of these degradation mechanisms in depth is necessary for their reliable use in space applications [93], [94], [95], [96]. High energy particles of radiation that come from the sun or from supernova explosions can cause a heavy error in logic devices when they bump into the semiconductor material. On striking such particles as protons and alpha particles create electron-hole pairs (EHPs), and they disrupt the p-n junction at the drain node. The resulting ionized carriers diffuse through the substrate to cause unwanted flow of current. A nondestructive phenomenon called single-event effect (SEE) can trigger temporary soft errors, which in many instances cause malfunctions of logic and memory circuits (Fig. 11). Such reliability problems caused by radiation are especially important in applications in space, defence, communications and autonomous systems for driving. Since the exposure to radiation particles is impossible to shield completely, the exposure is inevitable - military equipment used on the stratosphere, satellites and space stations used in the mesosphere, the use of digital systems from the troposphere to the Earth's surface. The partially surrounded channel structure of the FSFET gives little control over the formed EHPs by particle strikes and results in more unintended current compared to the NSFET. By contrast, the stacked nFET-pFET configuration of the CFET allows simultaneous particle interaction with both the nFET and pFET which enables their responses to counterbalance each other and thereby mitigating radiation-induced effects.

VI. CONCLUSION

The CFET architecture provides fantastic design flexibility for design implementation based on FinFETs, nanosheets (NSs), nanowires (NWs), or a combination of these (hybrid integration) between the bottom and top FETs; the way to future scaling and performance optimization in next-generation devices. A common gate is driving the vertically stacked p-/n-FETs, which will allow for a substantial decrease in device area, and provide a highly efficient approach for ultra-dense integration. The semiconductor industry is moving into the 'More Moore' era, spurred by schemes to integrate more and more circuits in 3D, thus transcending the limits of scaling excesses of traditional 2D- developments. While advanced packaging technologies have made 3D IC commercialization feasible, the microbumps and through-silicon vias bring large parasitic capacitances and extra area overhead, and ultimately limit 3D IC performance. Monolithic 3D integration has become an attractive option for next-generation 3D ICs; however, implementing monolithic 3D integration in silicon integrated circuits is hindered by the approach thermal budget over the top (which can negatively affect the performance of devices). To compensate for these shortcomings, new materials like graphene, carbon nanotubes and other 2D-semiconductors are being considered for incorporation into back end of silicon ICs. 2D semiconductors, such as transition metal dichalcogenides (TMDs), have gained tremendous attention in recent years as potential materials to extend Moore's Law beyond 2 nm node. TMD channel materials are highly robust to SCEs and have excellent carrier mobility at lower technology nodes than silicon.

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